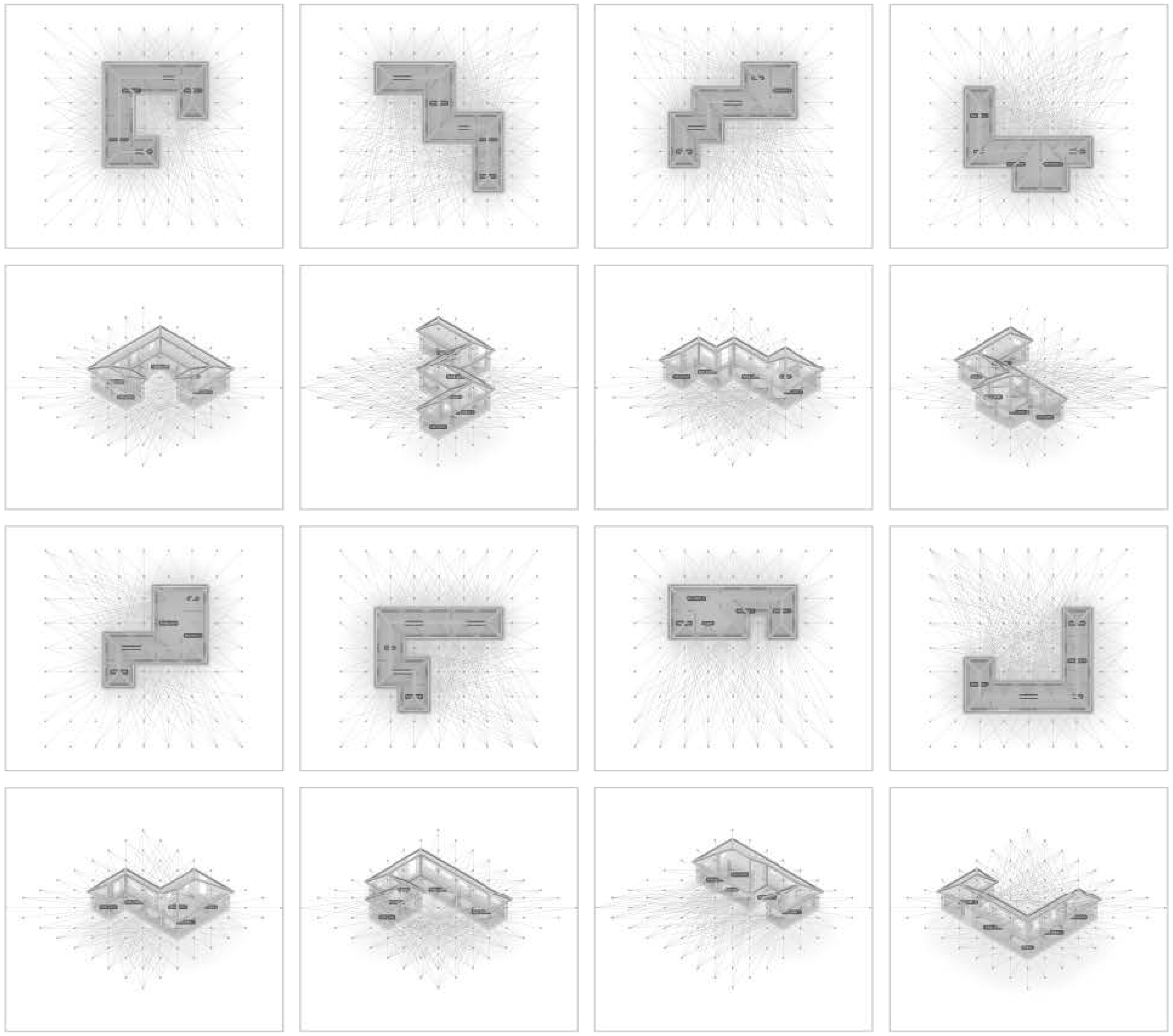
CHANGMIN JI

Architecture & Computational Design

01. Modular System (Object Oriented Architecture)
02. Design Alternative (Ecological & Sustainable Architecture)
03. Topography Generator (Site Modeling)

Modular System (Object-Oriented Architecture)

@ 2024/5th Year 1st Semester/Personal

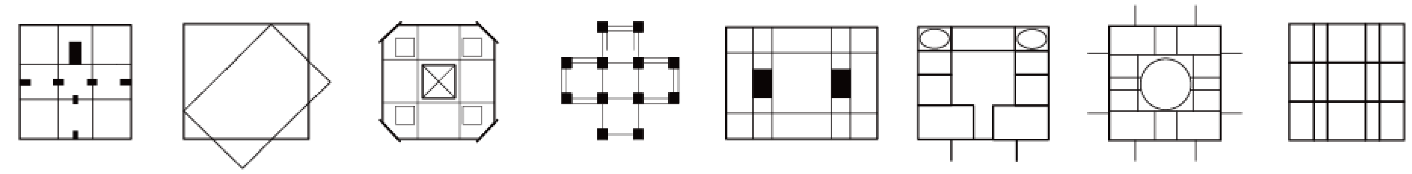


Modular System

Agenda

"What Kind of Architecture Should We Pursue?" The advancement of technology brings significant changes to human life, offering temporal, spatial, and economic surpluses. This progress becomes a driving force that opens new opportunities and possibilities for humanity. At this point in time, what new opportunities and possibilities does technological development offer to architecture? And what new opportunities and possibilities could it provide in the future?

From the perspective that "architecture is a system," we can discover these new opportunities and possibilities. Rudolf Wittkower identified geometric patterns in Palladio's eleven villas, strongly suggesting the existence of systems inherent in architecture. Structure, circulation, sequences, geometric patterns, and spatial hierarchies are all parts of the systems that coexist with architecture. Our future architecture must build upon this systemic approach, maximizing the benefits of technology to advance in a more innovative and sustainable direction.

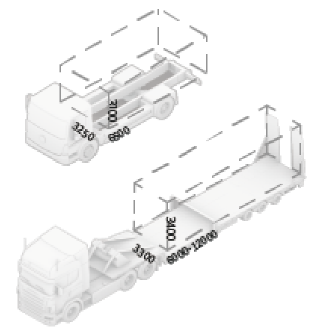


Temple of Solomon | HOUSE III | Exter Academy Library | Trenton Bath House | Villa Thiene | Villa Cornaro | Villa Rotonda | Geometrical Pattern of Palladio's Villas

Experiment

Unlike urban areas, the quality of residential living in rural areas is often inadequate. Despite the availability of standardized rural housing design plans, their simple floor configurations and limited layout types fail to adequately reflect the preferences of residents. Therefore, a modular system is proposed to enhance residential quality by accommodating user preferences.

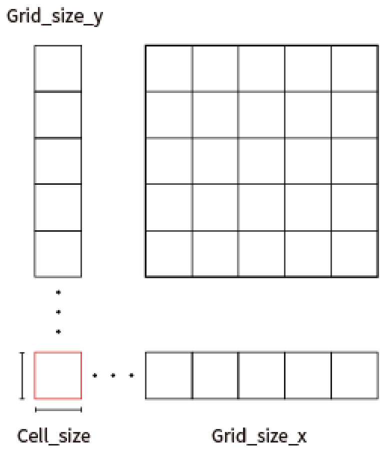
In rural areas, a modular system for single-family homes differs from urban contexts by reflecting and engaging with the surrounding environment, while also benefiting from fewer constraints and, most importantly, cost efficiency. Within this context, under the premise that "architecture is a system," modularity serves as a spatial unit and standard for organizing and understanding space, as well as a tool for architectural experimentation.



Experiment Procedure

Phase 1 System Setting	Phase 2 Space Organization	Phase 3 Window Planning	Phase 4 Roof Planning
Grid system Room system Start cell system	Connecting to each Module Assigning rooms Space hierarchy Setting door and entry position	Checking view sight Visualization view sight Setting window	Hip roof Setting hip roofs

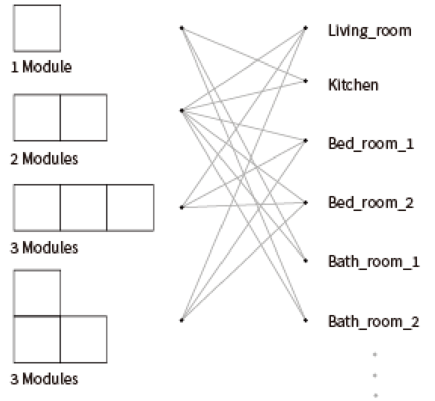
System Development



1.1 Grid system

Determine the size of the base cell and decide the number of cells to place along the x-axis and y-axis to define the grid size.

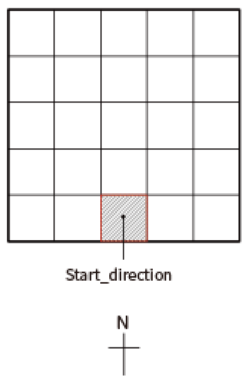
Parameter : Grid_size_x, Grid_size_y, Cell_size



1.2 Room system

Decide the types and numbers of rooms, and determine the size of each room by assigning a specific number of modules to each.

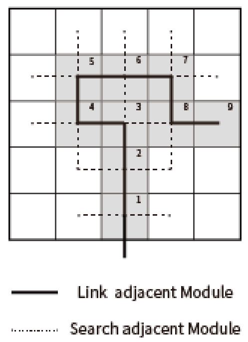
Parameter : Living_room, Kitchen, Bed_room, Bath_room



1.3 Start cell system

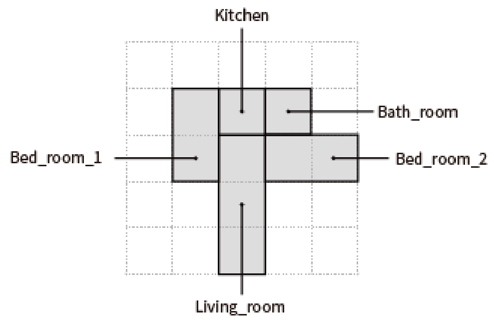
Choose one direction—east, west, south, or north—to determine the starting cell of the grid system.

Parameter : Start_direction : East, West, South, North



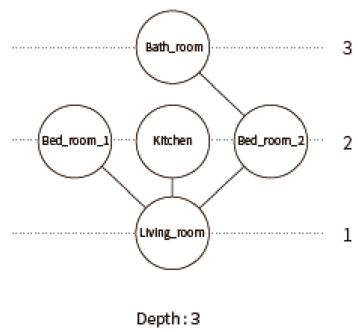
2.1 Connecting to each module

From the starting cell, explore adjacent cells, track the explored cells, and connect the modules along the explored path.



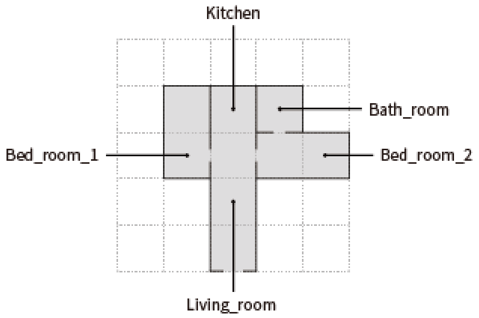
2.2 Assigning rooms

Referencing the values set in steps 1 and 2, assign the type and size to each room.



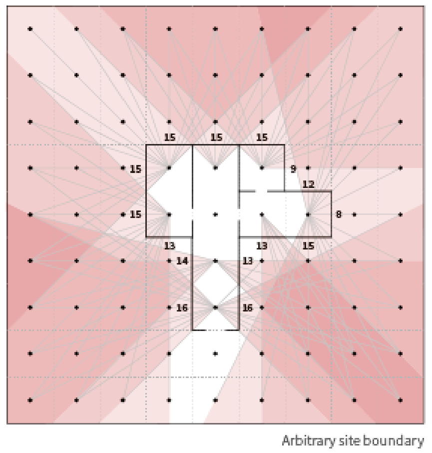
2.3 Space hierarchy

Arrange the spaces according to their hierarchy. For example, connect the living room and kitchen while separating Bedroom 1 and Bedroom 2.



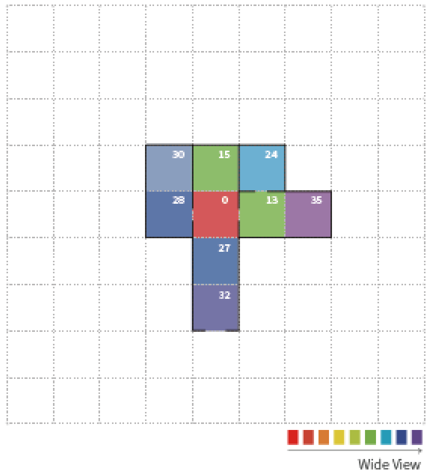
2.4 Setting door and entry position

Arrange each space based on the spatial hierarchy and assign doors accordingly. Ensure the entrance is accessible through the living room. When connecting the bathroom to other spaces, assign only one door to link it to a single space.



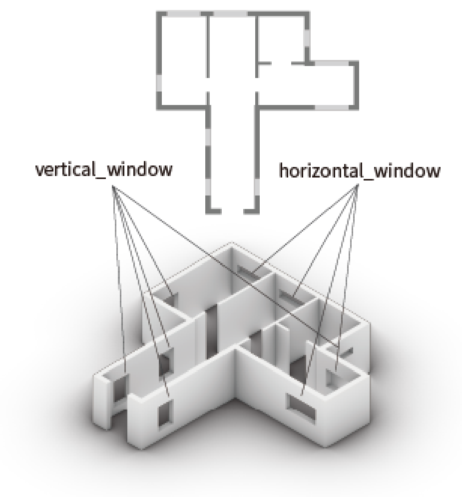
3.1 Checking view sight

Connect the points inside the module with those outside the module through the window locations to analyze visibility, and check the visibility score for each edge.



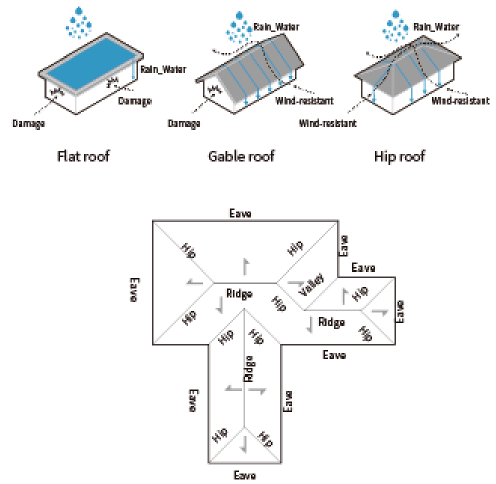
3.2 Visualization view sight

Evaluate the visual view score for each room and visualize the results.



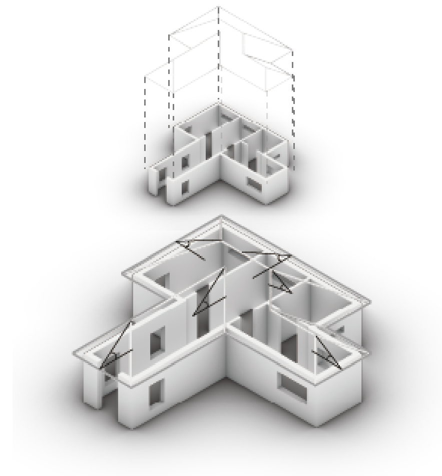
3.3 Setting window

Design vertical windows for the east and west sides, and horizontal windows for the north and south sides. Based on the visual view scores, determine whether each window is needed and adjust its size accordingly.



4.1 Hip roof

A hip roof, with its four sloping sides, offers better wind resistance compared to other roof types. Its design also allows water and snow to flow off easily, reducing the risks of flooding or snow load.

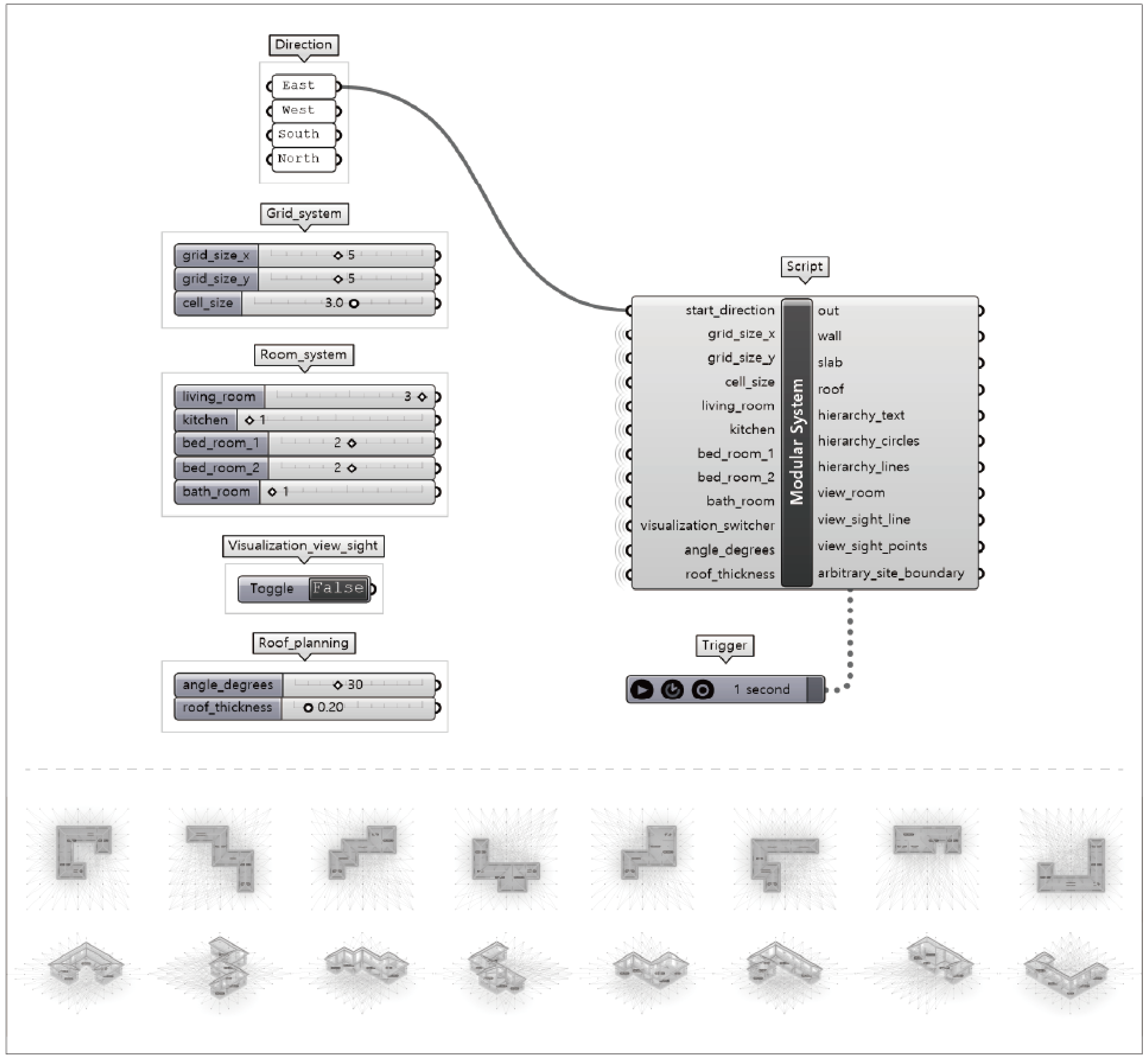


4.2 Setting hip roofs

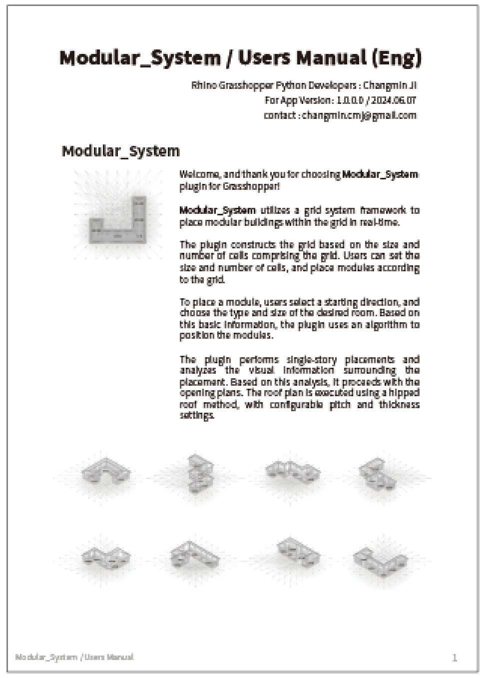
The hip roof accommodates structural integrity and design versatility, allowing it to adapt flexibly to various layout configurations.

Parameter : Eaves, Angle_degrees, Roof_thickness

Grasshopper & Python



Exhibition



Modular_System / Users Manual (Eng)

Rhino Grasshopper Python Developers : Changmin Ji
For App Version : 1.0.0.0 / 2024.06.07
contact : changmin.cmj@gmail.com

Modular_System

Welcome, and thank you for choosing **Modular_System** plugin for Grasshopper!

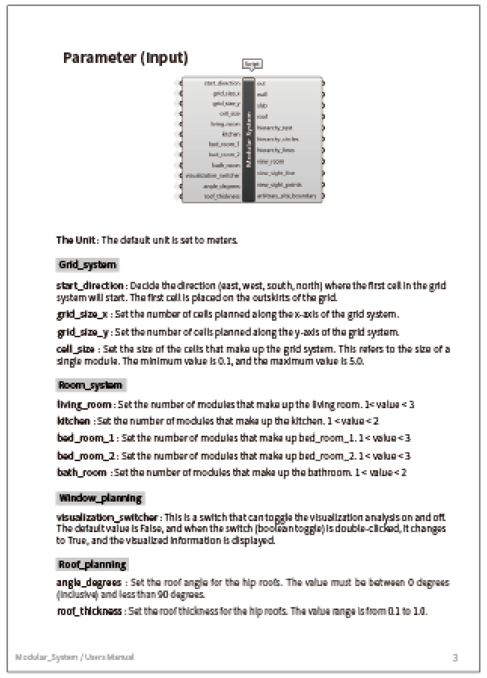
Modular_System utilizes a grid system framework to place modular buildings within the grid in real-time.

The plugin constructs the grid based on the size and number of cells comprising the grid. Users can set the size and number of cells, and place modules according to the grid.

To place a module, users select a starting direction, and choose the type and size of the desired room. Based on this basic information, the plugin uses an algorithm to position the modules.

The plugin performs single-story placements and analyzes the visual information surrounding the placement. Based on this analysis, it proceeds with the opening plans. The roof plan is executed using a hipped roof method, with configurable pitch and thickness settings.

Modular_System / Users Manual 1



Parameter (Input)

The Unit : The default unit is set to meters.

Grid_system

start_direction : Decide the direction (east, west, south, north) where the first cell in the grid system will start. The first cell is placed on the outskirts of the grid.

grid_size_x : Set the number of cells planned along the x-axis of the grid system.

grid_size_y : Set the number of cells planned along the y-axis of the grid system.

cell_size : Set the size of the cells that make up the grid system. This refers to the size of a single module. The minimum value is 0.1, and the maximum value is 5.0.

Room_system

living_room : Set the number of modules that make up the living room. 1< value < 3

kitchen : Set the number of modules that make up the kitchen. 1 < value < 2

bed_room_1 : Set the number of modules that make up bed_room_1. 1 < value < 3

bed_room_2 : Set the number of modules that make up bed_room_2. 1 < value < 3

bath_room : Set the number of modules that make up the bathroom. 1 < value < 2

Window_planning

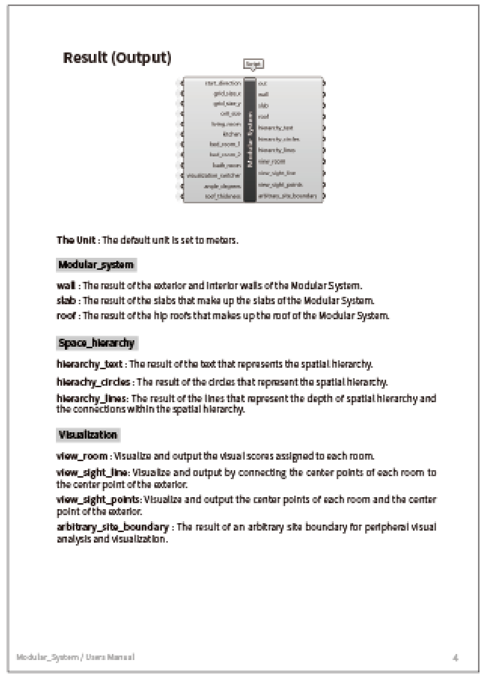
visualization_switcher : This is a switch that can toggle the visualization analysis on and off. The default value is False, and when the switch (boolean toggle) is double-clicked, it changes to True, and the visualized information is displayed.

Roof_planning

angle_degrees : Set the roof angle for the hip roofs. The value must be between 0 degrees (inclusive) and less than 90 degrees.

roof_thickness : Set the roof thickness for the hip roofs. The value range is from 0.1 to 1.0.

Modular_System / Users Manual 3



Result (Output)

The Unit : The default unit is set to meters.

Modular_system

wall : The result of the exterior and interior walls of the Modular System.

slab : The result of the slabs that make up the slabs of the Modular System.

roof : The result of the hip roofs that makes up the roof of the Modular System.

Space_hierarchy

hierarchy_text : The result of the text that represents the spatial hierarchy.

hierachy_circles : The result of the circles that represent the spatial hierarchy.

hierarchy_lines: The result of the lines that represent the depth of spatial hierarchy and the connections within the spatial hierarchy.

Visualization

view_room : Visualize and output the visual scores assigned to each room.

view_sight_line: Visualize and output by connecting the center points of each room to the center point of the exterior.

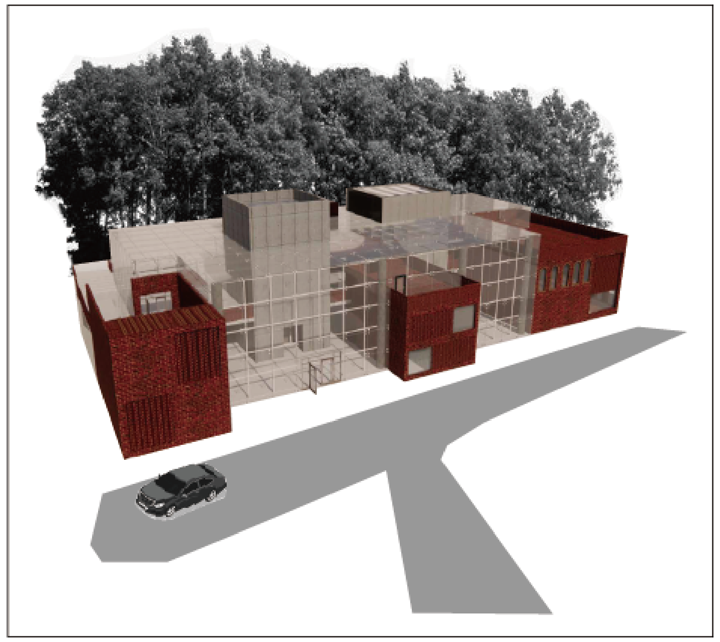
view_sight_points: Visualize and output the center points of each room and the center point of the exterior.

arbitrary_site_boundary : The result of an arbitrary site boundary for peripheral visual analysis and visualization.

Modular_System / Users Manual 4

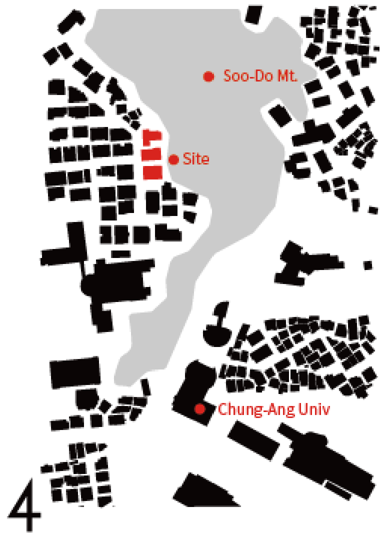
Design Alternative (Ecological & Sustainable Architecture)

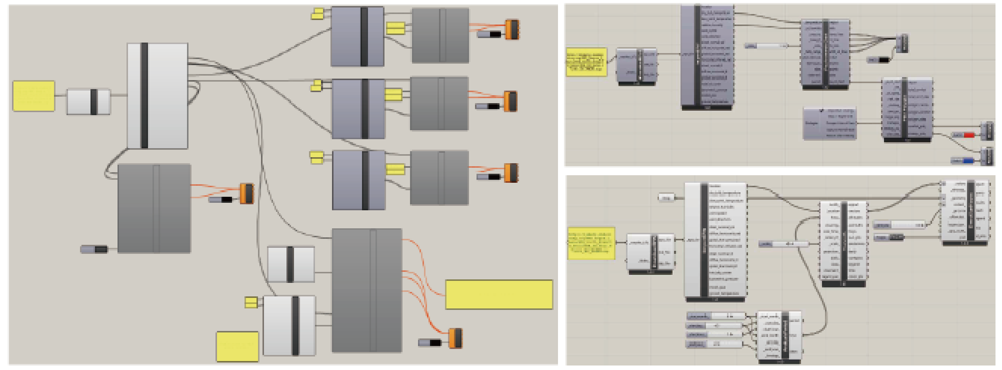
@ 2022/3th Year 1st Semester/Teamwork



Before

After

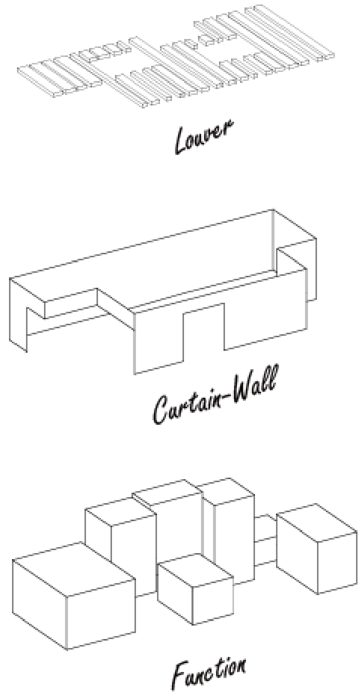




Building Info

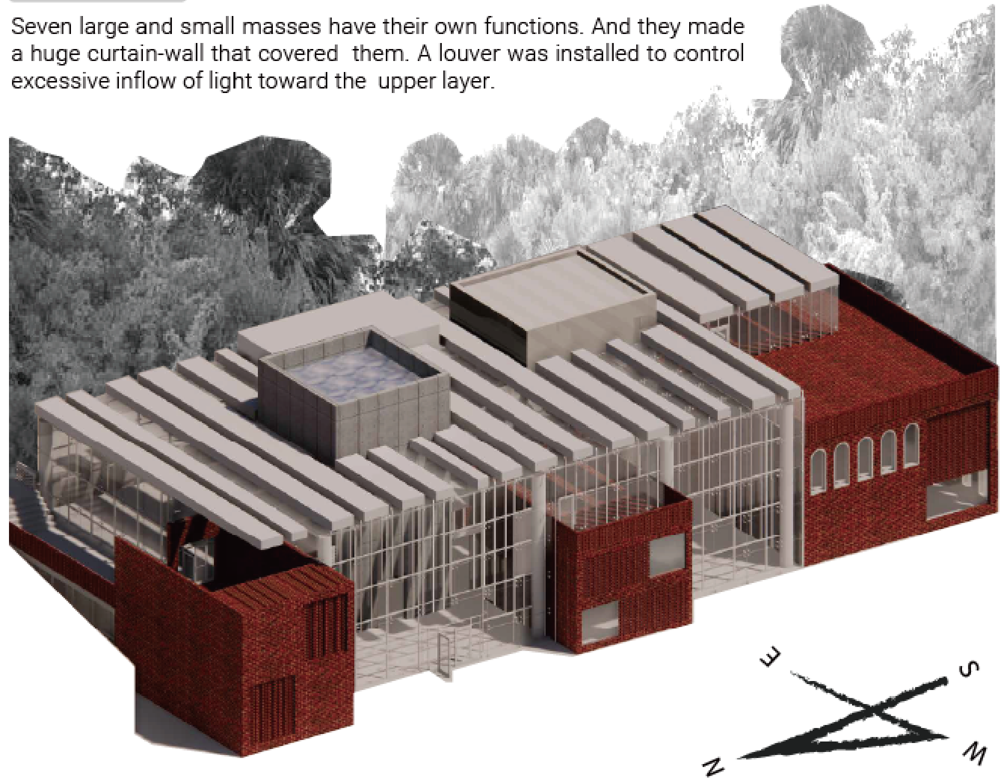
Location_ 21,23,27 Maebong-ro 4ga-gil, Dongjak-gu, Seoul **Program_** Library

Area_ 1,180㎡ **Floor_** 1F-3F **Building Type_** Internally Dominated



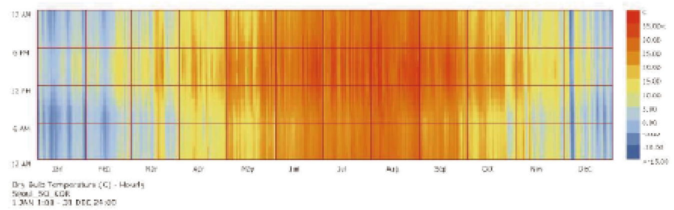
Previous Design

Seven large and small masses have their own functions. And they made a huge curtain-wall that covered them. A louver was installed to control excessive inflow of light toward the upper layer.

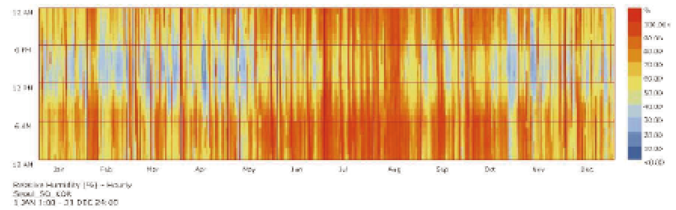


Site Analysis

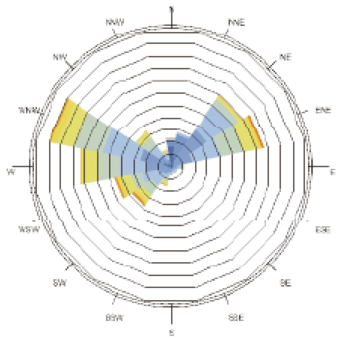
1. DBT & RH



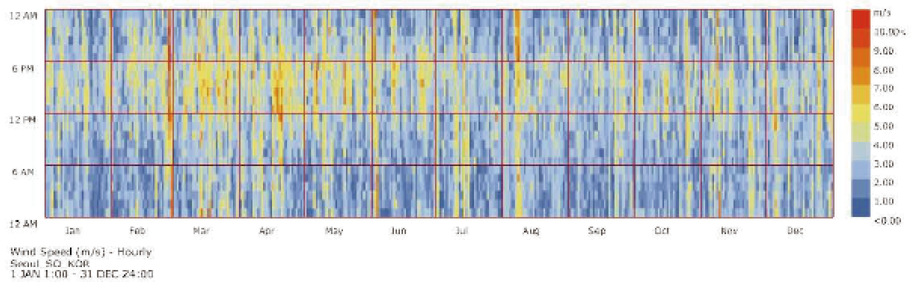
1



2



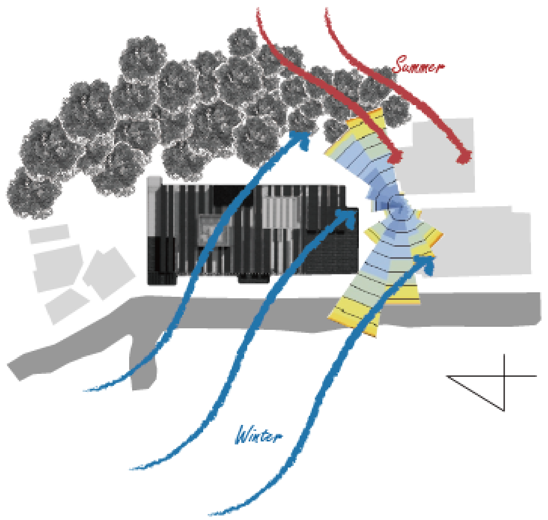
1



2

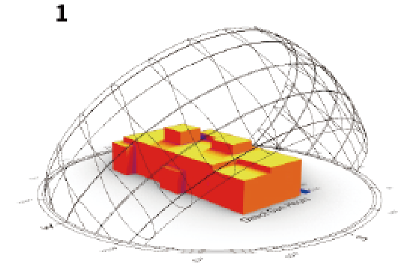
2. Wind Path

The wind utilization plan for eco-friendly design can be divided by season. In winter in December, January, and February, the main wind direction was the northwest early wind and the minor wind direction, the westwind, accounted for 35-40% of the total.



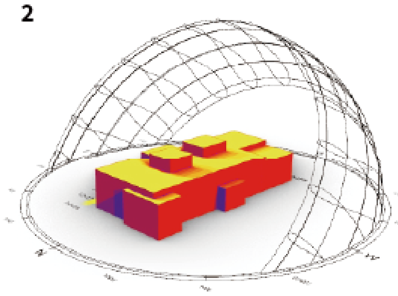
Summer Solstice

1



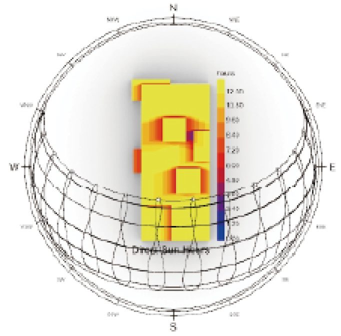
WS

2



NW

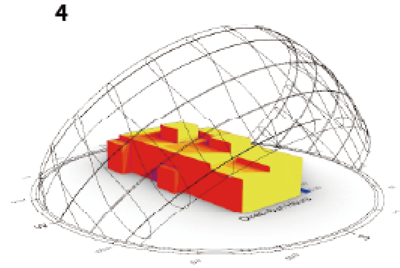
3



Top view

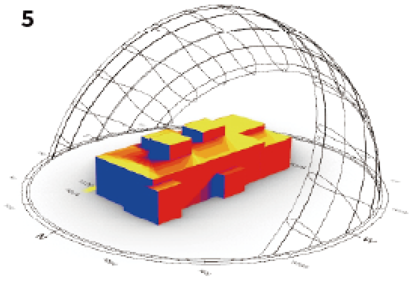
Winter Solstice

4



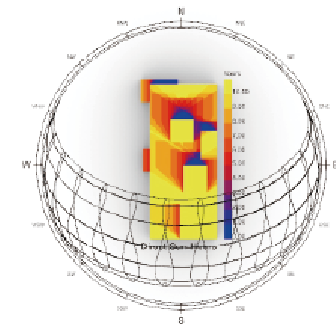
WS

5



NW

6

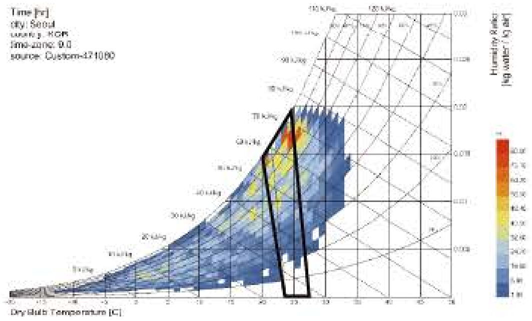


Top view

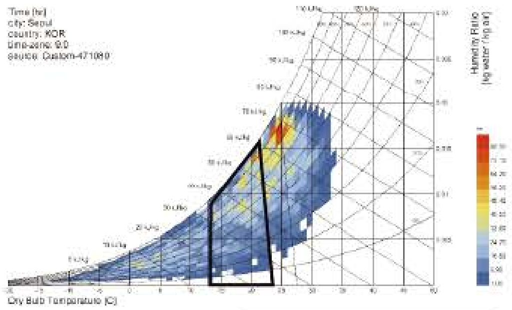
3. Sun Path & Sunlight Hour Analysis

It shows how much sunlight the building receives when the sun is up.

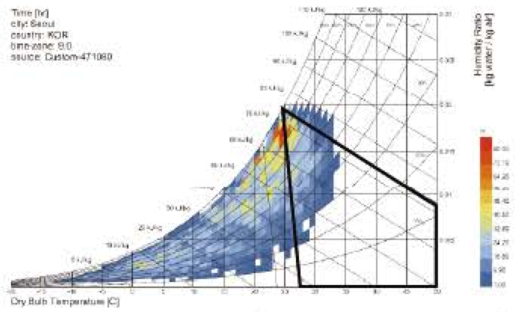
4. Psychrometric chart



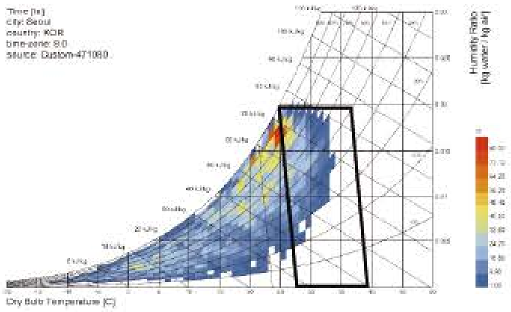
1. Default Comfort Zone



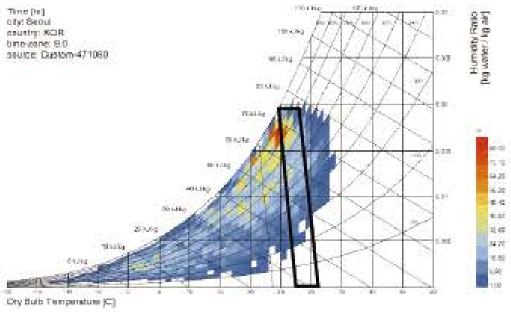
2. Capture Internal Heat



3. Evaporative Cooling

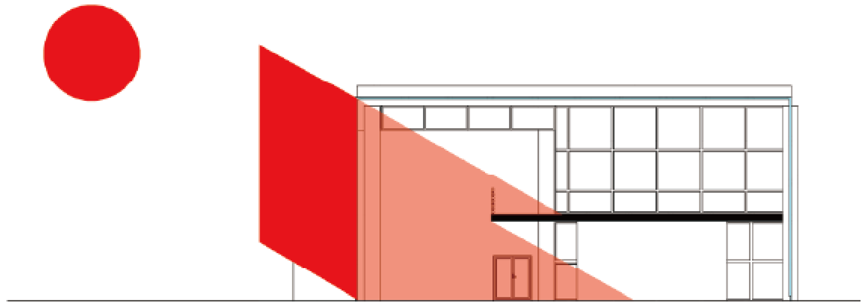


4. Thermal Mass+Ventilation



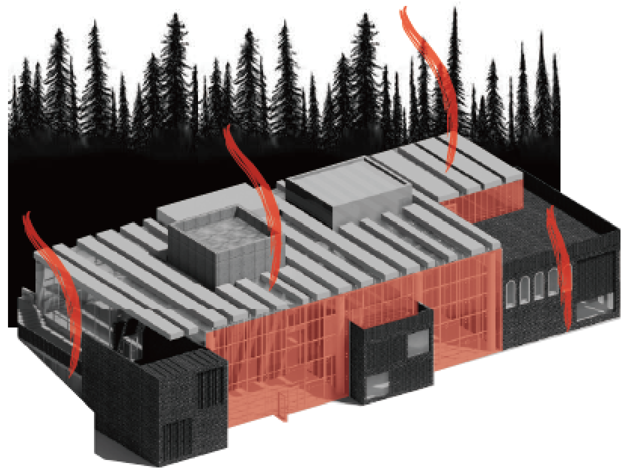
5. Occupant use of fans

Problem



1. Lighting & Shading

Currently, only wooden louver installed on the ceiling remains. It can't stop the sun-light coming in from the west.

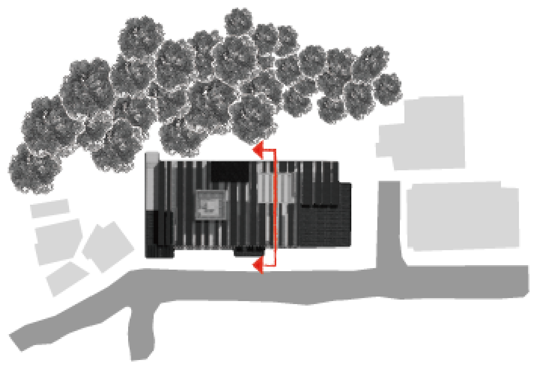


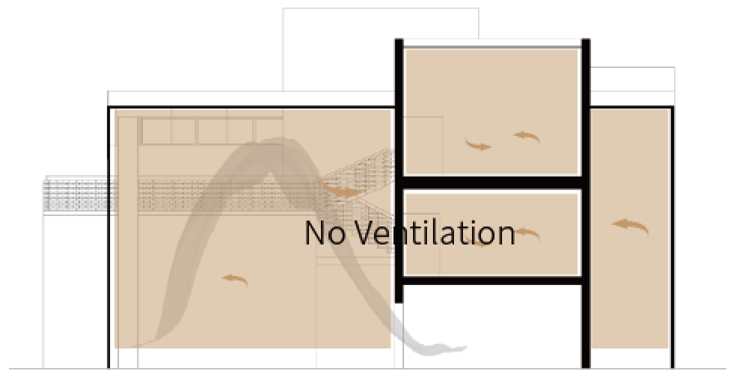
2. Engergy Efficiency

A huge curtain wall made of thin glass loses a lot of energy. In summer, it makes the interior hot like a huge glass green-house. In winter, heating efficiency drops sharply, making the room cold. Countermeasures are needed to create a pleasant interior while maintaining the design concept.

3. Ventilation

The design previously considered has few windows, so ventilation is not good. When viewed from a cross-sectional view, the interior works almost like a closed system. The air is barely allowed in except for a few windows and doors.



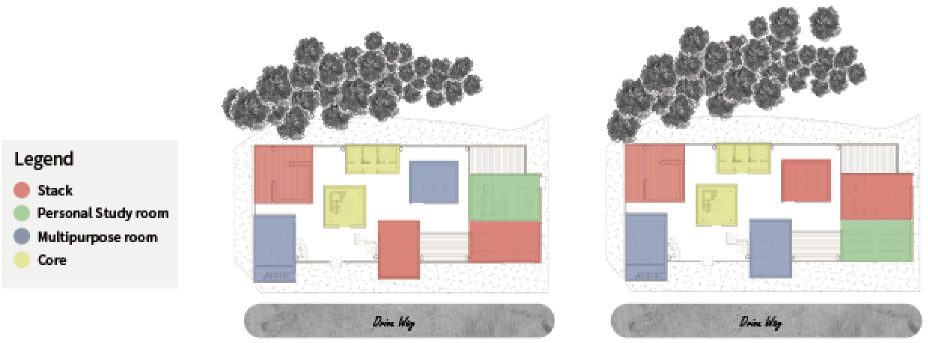


Solution

1. Re-Zonning

Through this eco-friendly study, we found out about the lighting in the field. As a result, we found that traditional zoning was not designed to fit the library design method. It is placed behind with books vulnerable to light (direct sunlight) so that books can be stored in a shaded place.

Before
After

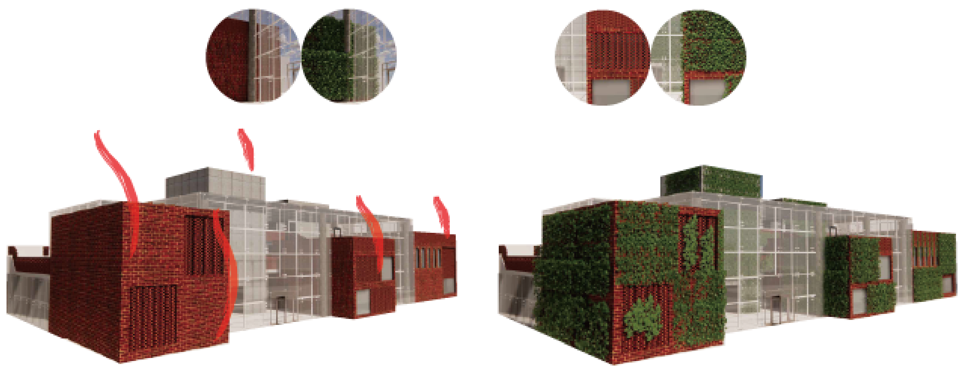


2. Greening

Frequent exposure to the sun increases the demand for energy significantly in summer as people use air conditioning and cooling systems. Wall greening helps save energy for buildings as well as walls, and in the case of transpiration plants, you can actually reduce the temperature of the room anywhere by cooling the surrounding environment.

In Door - Transpiration
(Similar to Evaporative cooling)

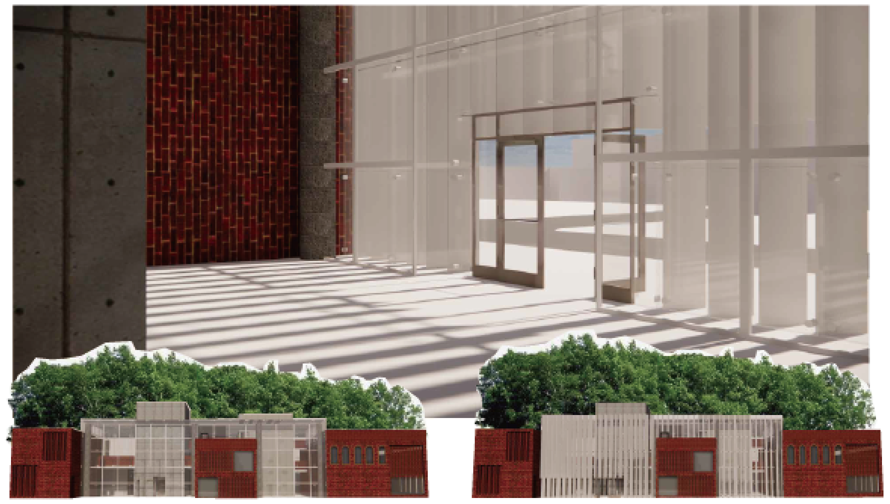
Out Door - Shading
(Similar to Window Blind)



Before
After

3. Making Shade (Vertical fin)

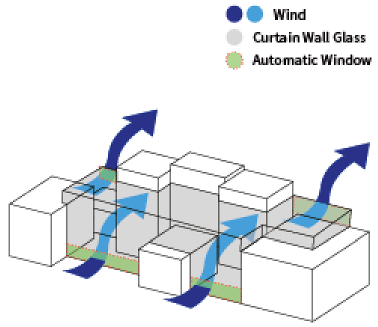
Currently, only wooden louver installed on the ceiling remains. It can't stop the sunlight coming in from the west.

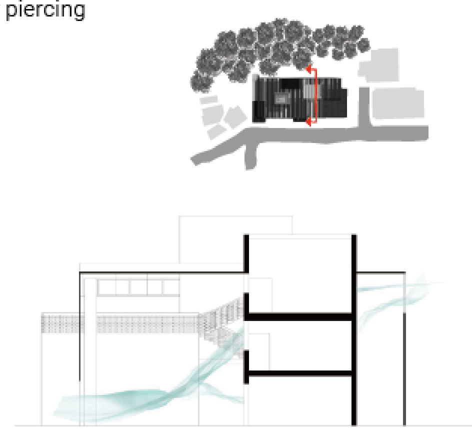


Before
After

4. Dog Trot

In order to use efficient ventilation using Dog trot, windows were made on curtain wall glass. This window can be opened and closed automatically even out of reach. The principle of dog trot was applied by piercing the lower front part and the upper back part.

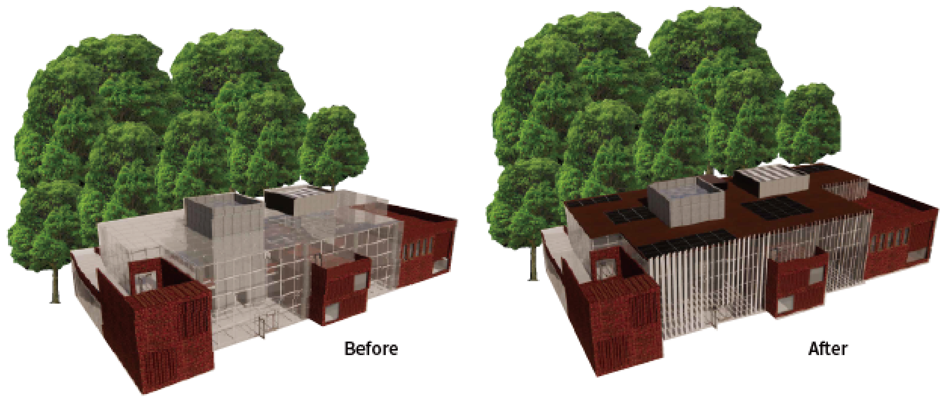




Section

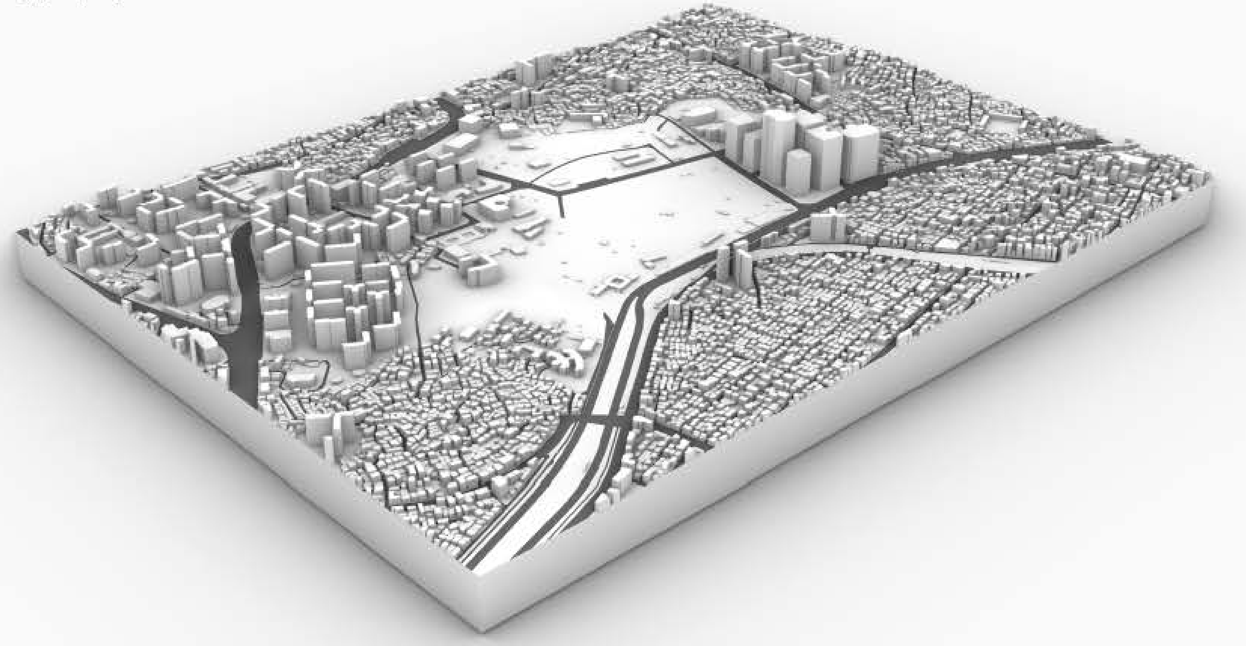
5. PV Panel

Install solar power generators on the roof using good lighting conditions. It was placed on a scale similar to the mass scale to match the initial design concept.



Topography Generator (Site modeling)

@2024/Toy project/Personal



Site Modeling Plan

Agenda

In a digital design environment, unnecessary and repetitive tasks inevitably arise during architectural design. These tasks consume time and energy that could otherwise be invested in intellectual activities. The Topography Generator in a digital design environment aims to partially address these challenges.

Site modeling serves as the foundation for the 3D environment during the early stages of design. It enables easier three-dimensional analysis of contextual elements such as skylines and traffic in the surrounding environment, while also supporting visualization tasks like diagrams and renderings.

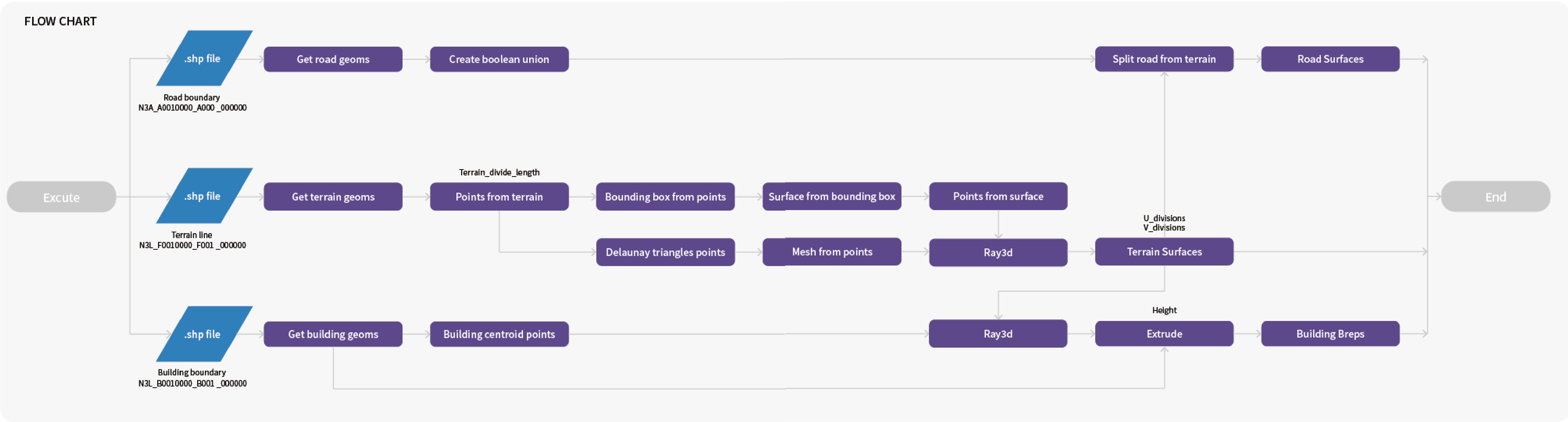
The Topography Generator utilizes the SHP files from the national geographic information map's continuous numeric maps to separate terrain, building, and road data and generate 3D models. Users simply need to input the folder path and precision level, and the tool automatically performs the tasks, enabling fast and efficient site modeling.

Input

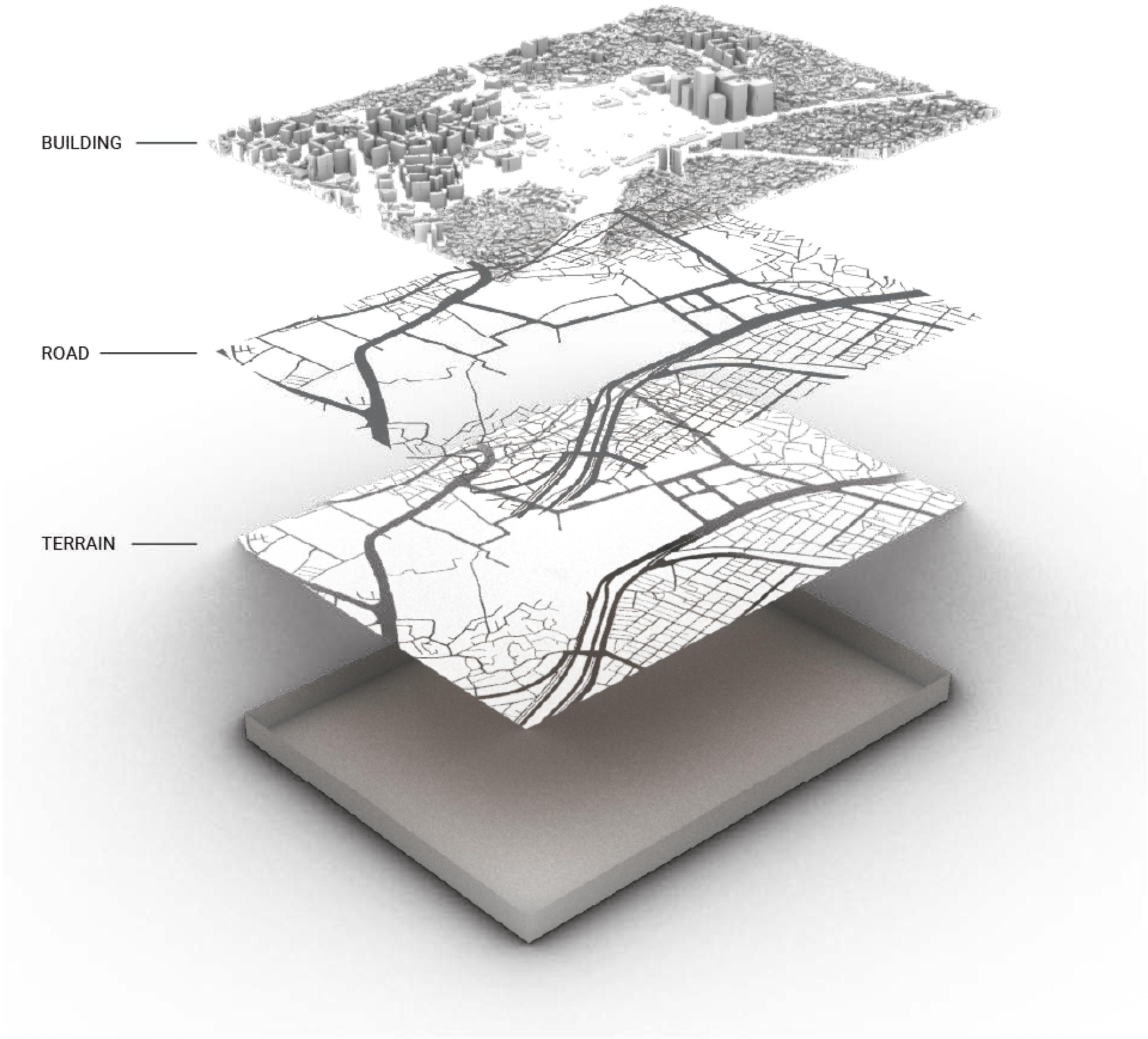
- **Continuous Numerical Map Folder Path**
  - Road Boundary / N3A_A0010000_A000 _000000.shp
  - Terrain Line / N3L_F0010000_F001 _000000.shp
  - Building Boundary / N3L_B0010000_B001 _000000.shp
- **Precision**
  - terrain_divide_length
  - u_divisions
  - v_divisions
- **Building height**
  - height

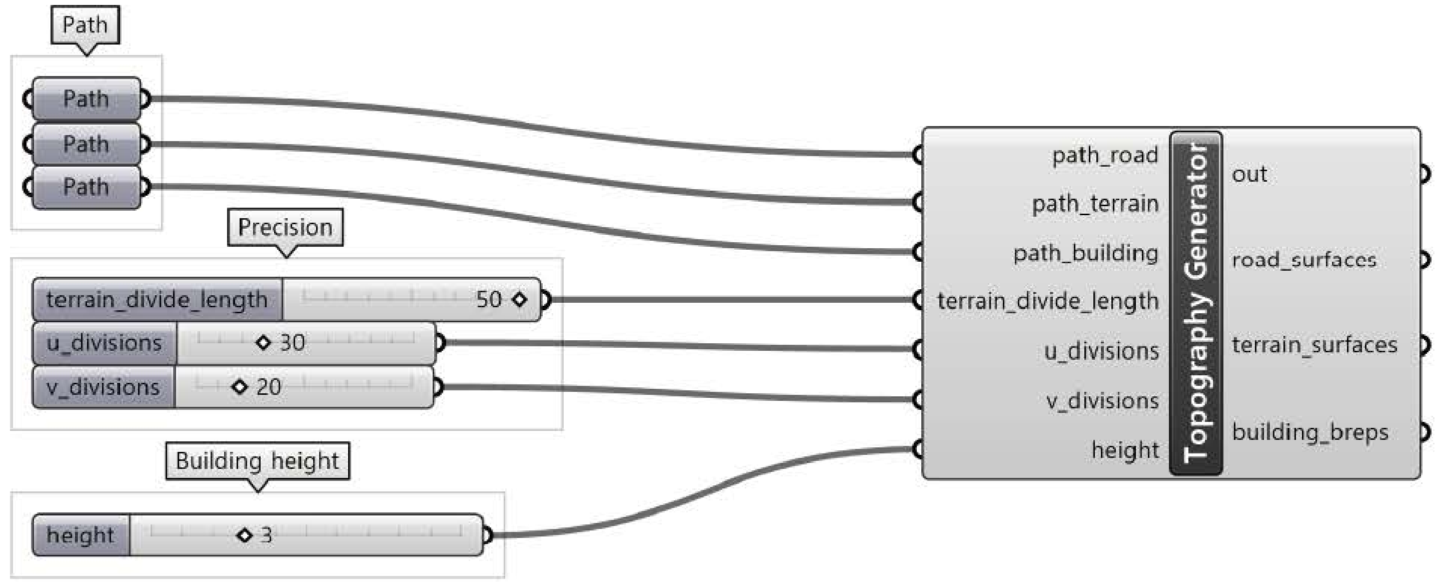
Output

- road_surfaces
- terrain_surfaces
- building_breps



FLOW CHART
Excute
.shp file
Road boundary
N3A_A0010000_A000 _000000
.shp file
Terrain line
N3L_F0010000_F001 _000000
.shp file
Building boundary
N3L_B0010000_B001 _000000
Get road geoms
Create boolean union
Split road from terrain
Road Surfaces
Get terrain geoms
Terrain_divide_length
Points from terrain
Bounding box from points
Surface from bounding box
Points from surface
Delaunay triangles points
Mesh from points
Ray3d
U_divisions
V_divisions
Terrain Surfaces
Get building geoms
Building centroid points
Ray3d
Height
Extrude
Building Breps
End


BUILDING
ROAD
TERRAIN


Path
Path
Path
Path
Precision
terrain_divide_length
50
u_divisions
30
v_divisions
20
Building height
height
3
path_road
path_terrain
path_building
terrain_divide_length
u_divisions
v_divisions
height
Topography Generator
out
road_surfaces
terrain_surfaces
building_breps